Tally Hardware Manual

Document TM-HW-07, revision E, August 2026. Applies to Tally Tag TT-200, Tally Tag TT-200C and Tally Hub TH-4.

1. System overview

A Tally installation has three parts. **Tally Tags** are battery-powered load sensors mounted under shelf sections; they measure the weight of the stock on the shelf. Tags send readings over a sub-GHz radio link to a **Tally Hub**, a gateway that forwards the data to **Tally Cloud**, where stock levels, alerts and reports are available in the dashboard and through the REST API.

2. Tally Tag TT-200

- Measuring range: 0 to 150 kg per tag, with a resolution of 20 g.
- Power: 2 × CR123A lithium batteries.
- Battery life: about 28 months at the default reporting interval of 5 minutes, and about 11 months at a 1-minute reporting interval.
- Operating temperature: −20 °C to +50 °C.
- Radio: 868 MHz in the EU and 915 MHz in the US, with an indoor range of up to 60 m in line of sight.
- Ingress protection: IP54.

The **TT-200C** is the cold-storage variant for freezers and chilled rooms. It operates from −35 °C to +10 °C, is rated IP66, and uses the same batteries. At temperatures below −20 °C, expect battery life to drop by roughly 30%.

3. Tally Hub TH-4

- Capacity: up to 250 tags per hub. For reliable performance we recommend no more than 200.
- Uplink: Ethernet with PoE+ as the primary connection, with automatic LTE Cat-M1 fallback when Ethernet is unavailable.
- Offline buffer: if the hub loses its connection to Tally Cloud, it stores up to 72 hours of tag readings and uploads them automatically when the connection returns.
- Mounting height: at least 2 m above the floor, as central as possible within the area it serves.

4. Installation

1. Mount the hub centrally, no more than 60 m from the furthest tag, at least 2 m high.
2. Connect Ethernet (PoE+) or insert the LTE SIM. The hub LED turns solid white when it has booted and is online.
3. Claim the hub: in Tally Cloud, open Sites → Hubs → Add hub and enter the 12-character claim code printed on the hub label.
4. Pair each tag: press and hold the tag button for 5 seconds until its LED blinks blue. The tag joins the nearest hub that is in pairing mode. A hub stays in pairing mode for 10 minutes after you click "Pair tags" in the dashboard.
5. Calibrate each tag: with the shelf empty, run "Tare" in the dashboard, then place the 5 kg reference weight included in the installation kit and run "Calibrate".

A tag must be recalibrated whenever it is moved to a different shelf. Replacing batteries does not require recalibration.

5. LED status codes — Tally Tag

LED pattern	Meaning	What to do
Blinking blue	Pairing mode	Wait until pairing completes
Solid green for 3 seconds	Paired successfully	Nothing
Amber blink every 30 seconds	Battery below 15%	Replace both batteries within 2 weeks
Blinking red	Cannot reach the hub	Check the distance to the hub and for obstacles; check the hub is online
Alternating red and amber	Load cell fault	Contact support to start an RMA

6. LED status codes — Tally Hub

LED pattern	Meaning	What to do
Solid white	Booted and online	Nothing
Pulsing cyan	Firmware update in progress	Do not unplug the hub
Solid amber	Online through the LTE fallback	Check the Ethernet connection
Blinking red	No uplink at all	Check cabling, PoE+ power and the SIM

7. Firmware

Hubs install firmware updates automatically during the maintenance window of 02:00 to 04:00 in the site's local time. The hub then updates its tags over the air, one tag at a time, so tag updates for a large site can take several nights. Current versions are hub firmware 5.8.2 and tag firmware 2.4.0.

8. Factory reset

To factory-reset a hub, press and hold the recessed reset pin for 15 seconds until the LED flashes magenta three times. A reset hub must be claimed again in Tally Cloud, but its paired tags reconnect automatically within about 10 minutes. To reset a tag, remove one battery, hold the button, and reinsert the battery while holding the button for 10 seconds.

9. Safety and disposal

Use only CR123A lithium batteries. Do not mix old and new batteries. Dispose of used batteries and devices according to local e-waste regulations; in the EU, devices can be returned to Brindlewave free of charge for recycling.